Model Answers.

Name: Roll Number:

EE2001 Quiz I

Monday, 19/02/18

Closed book, 50 mins, 20 points

Remember...

- *Make reasonable assumptions when in doubt. However, you should show all steps in additional sheets and justify assumptions for full credit.* ***Provide final answers ONLY in the space provided.***

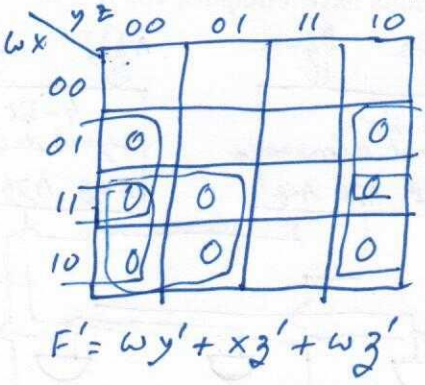
1. In a particular number system of base r, the roots of the equation $5x^2 + 50x + 125 = 0$ are given by $x = -8$ and $x = -5$. The value of r is 13. (2 pts)

$(x+8)(x+5) = 0 \Rightarrow x^2 + 13x + 40 = 0 \Rightarrow 5x^2 + 65x + 200 = 0.$

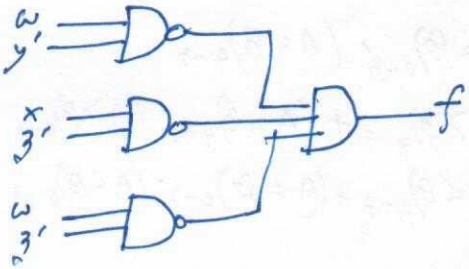
$5r = 65 \Rightarrow r = 13$; Also, $1 \times 13^2 + 2 \times 13 + 5 = 200$. (Confirmed).

2. Simplify the following Boolean function and implement the same using NAND-AND gates:

$F(w, x, y, z) = \Pi(4, 6, 8, 9, 10, 12, 13, 14)$ (3 pts)



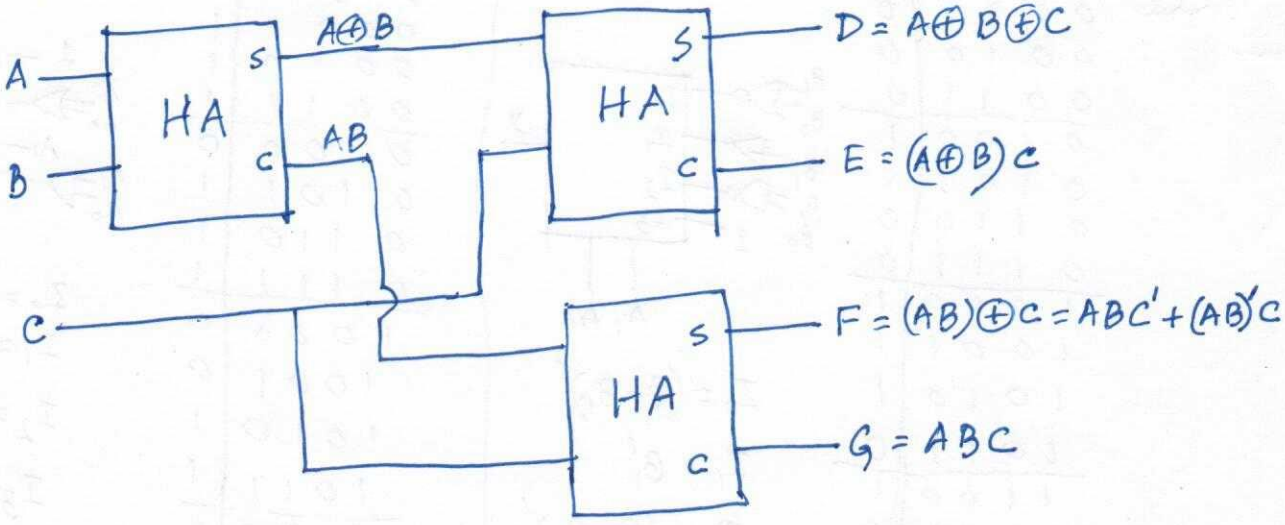
$F' = wy' + xz' + wz'$



3. Implement the following Boolean functions using ONLY three half-adder circuits: (4 pts)

(i) D = A ⊕B⊕C (ii) E = A'BC + AB'C (iii) F = ABC' + (A' + B')C (iv) G = ABC

S.C (A⊕B) C AB⊕C C.C



$G_i = A_i B_i$ $\quad S_i = P_i \oplus C_i$

$P_i = A_i \oplus B_i$

$C_i = G_{i-1} + P_{i-1} C_{i-1}$

XOR → 20ns

Other gates → 10ns

4. (a) Obtain Boolean expressions for the carry bits (C_1, C_2, C_3, C_4) of a 4-bit carry look ahead adder.

$C_1 = G_0 + P_0 C_0$

$C_2 = G_1 + P_1 G_0 + P_1 P_0 C_0$

$C_3 = G_2 + P_2 G_1 + P_2 P_1 G_0 + P_2 P_1 P_0 C_0$

$C_4 = G_3 + P_3 G_2 + P_3 P_2 G_1 + P_3 P_2 P_1 G_0 + P_3 P_2 P_1 P_0 C_0$

G → 20 $G_i = A_i B_i$

$P_i = A_i \oplus B_i$

20ns → P

10ns → $P_3 P_2 P_1 P_0 C_0$

10ns → Adding carry

20ns → Sum

(b) Assuming that the exclusive-OR gate has a propagation delay of 20 ns and all other gates have propagation delay of 10 ns, the total propagation delay for this 4-bit adder is .60 ns.

1 Ex-OR for generating G's and P's + 2 Gates for generating carries + 1 Ex-OR for Sum and carry

(c) If two such 4-bit adders are connected to realize an 8-bit adder, the total propagation delay is .....80 ns. (1+2+2 pts)

For the CLA for higher order bits, C_4 is available after 40 ns.

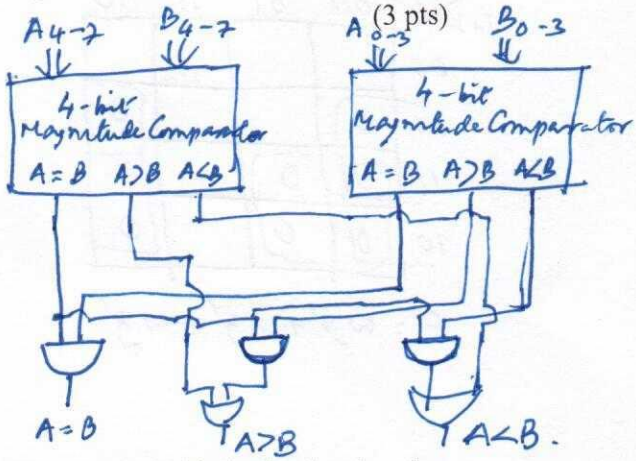
Delay = 40 ns + 20 ns (to generate all C's) + 20 ns for Sum o. tputs.

5. Design and draw the circuit of an 8-bit magnitude comparator realized using two 4-bit magnitude comparators and a few gates. The magnitude comparators have 3 outputs, viz. A = B, A > B and A < B. (3 pts)

$(A=B) = (A=B)_{0-3} \cdot (A=B)_{4-7}$

$(A>B) = (A>B)_{4-7} + (A=B)_{4-7} \cdot (A>B)_{0-3}$

$(A<B) = (A<B)_{4-7} + (A=B)_{4-7} \cdot (A<B)_{0-3}$

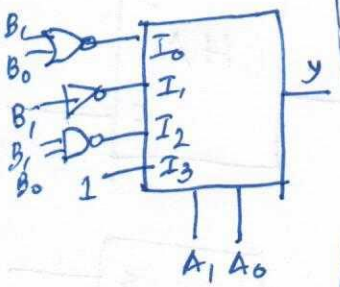


6. Two 2-bit binary numbers A = A_1A_0 and B = B_1B_0 are fed into a 4-bit logic circuit, whose output is 1 only if A ≥ B. Design and draw the circuit using one 4 x 1 MUX and some gates. (3 pts)

A_1A_0	B_1B_0	Y
0 0	0 0	1
0 0	0 1	0
0 0	1 0	0
0 0	1 1	0
0 1	0 0	1
0 1	0 1	1
0 1	1 0	0
0 1	1 1	0
1 0	0 0	1
1 0	0 1	1
1 0	1 0	1
1 0	1 1	0
1 1	0 0	1
1 1	0 1	1
1 1	1 0	1
1 1	1 1	1

OR

B_1B_0	A_1A_0	Y
0 0	0 0	1
0 0	0 1	1
0 0	1 0	1
0 0	1 1	1
0 1	0 0	0
0 1	0 1	1
0 1	1 0	1
0 1	1 1	1
1 0	0 0	0
1 0	0 1	0
1 0	1 0	1
1 0	1 1	1
1 1	0 0	0
1 1	0 1	0
1 1	1 0	0
1 1	1 1	1

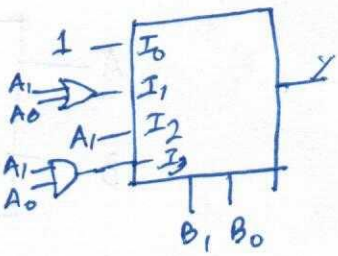


$I_0 = (B_1 + B_0)'$

$I_1 = B_1'$

$I_2 = (B_1 \cdot B_0)'$

$I_3 = 1$



$I_0 = 1$

$I_1 = A_1 + A_0$

$I_2 = A_1$

$I_3 = A_1 \cdot A_0$

Name: _____ Roll Number: _____

EE2001 Quiz II

Monday, 02/04/18

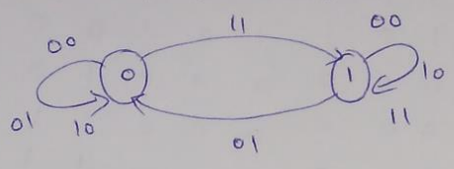
Closed book, 50 mins, 20 points

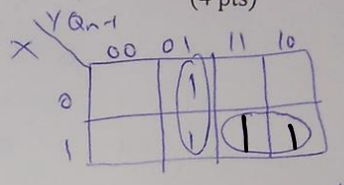
Remember...

- *Make reasonable assumptions when in doubt. However, you should show all steps and justify assumptions for full credit.* ***Provide final answers ONLY in the space provided.***

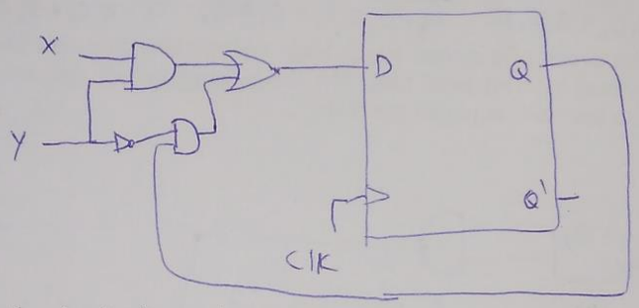
1. Draw the state diagram for a Flip-Flop which has two inputs X and Y and two outputs Q and Q'. When Y = 0, the output Q_n is equal to the previous state Q_{n-1} and when Y =1, Q_n = X. Realize the Flip-Flop using a D Flip-Flop and some gates. (4 pts)

X	Y	Q_{n-1}	Q_n
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

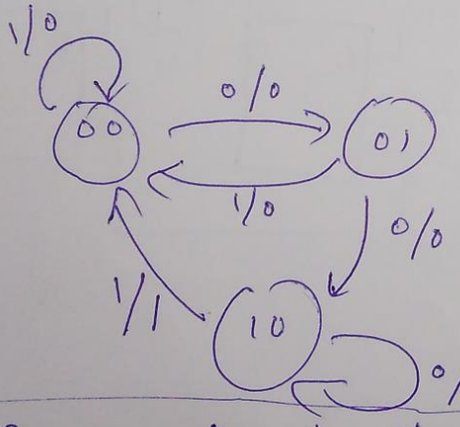


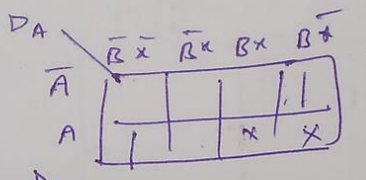


$Q_n = XY + Q_{n-1}Y'$

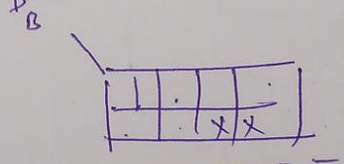


2. Design and draw the circuit of a synchronous single input single output system such that the output Z = 1 with every occurrence of an input 1 following a string following a string of at least two consecutive 0 inputs. At all other times, the output Z = 0. (4 pts)



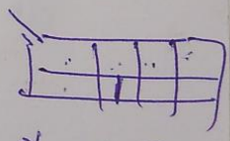


$D_A = \bar{x}(A+B)$



$\bar{A}\,\bar{B}\,\bar{x}$

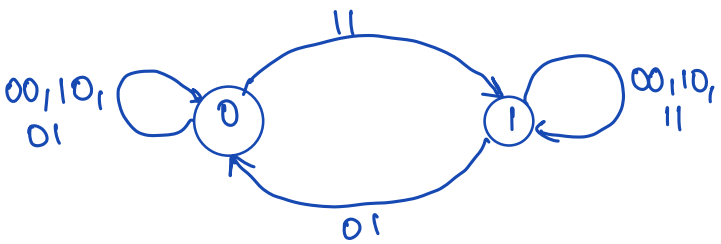
o/p



o/p = $A\bar{B}x$.

A	B	x	A^+	B^+	o/p	D_A	D_B
0	0	0	0	1	0	0	1
0	0	1	0	0	0	0	0
0	1	0	1	0	0	1	0
0	1	1	0	0	0	0	0
1	0	0	1	0	0	1	0
1	0	1	0	0	1	0	0
			x	x		x	x

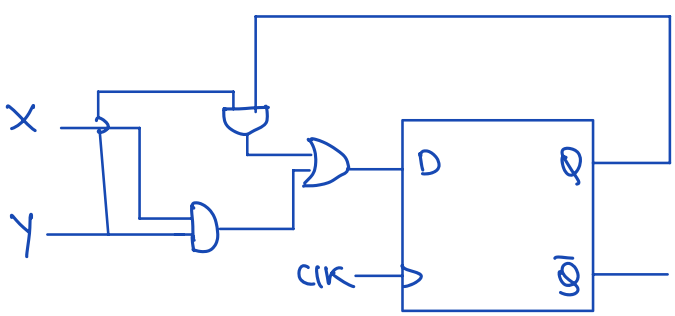
1. Draw the state diagram for a Flip-Flop which has two inputs X and Y and two outputs Q and Q'. When Y = 0, the output Q_n is equal to the previous state Q_{n-1} and when Y =1, Q_n = X. Realize the Flip-Flop using a D Flip-Flop and some gates. (4 pts)



X	Y	Q	Q^+
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

X \ YQ	00	01	11	10
0		1		
1		1	1	1

$Q^+ = \bar{Y}Q + XY$



2. Design and draw the circuit of a synchronous single input single output system such that the output Z = 1 with every occurrence of an input 1 following a string following a string of at least two consecutive 0 inputs. At all other times, the output Z = 0. (4 pts)

S_0 → previous input not zero

S_1 → last input was 0 (only once)

S_2 → last input was 0 (twice)

$S_3 \rightarrow$ last input was 1 (...001)

P.S	x	N.S	y
S_0	0	S_1	0
S_0	1	S_0	0
S_1	0	S_2	0
S_1	1	S_0	0
S_2	0	S_2	0
S_2	1	S_3	1
S_3	0	S_1	0
S_3	1	S_0	0

Q_1	Q_0	x	Q_1^+	Q_0^+	y
0	0	0	0	1	0
0	0	1	0	0	0
0	1	0	1	0	0
0	1	1	0	0	0
1	0	0	1	0	0
1	0	1	1	1	1
1	1	0	0	1	0
1	1	1	0	0	0

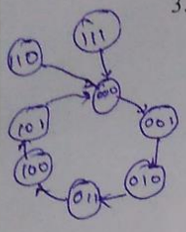
$y = Q_1\overline{Q_0}x$

$Q_1^+ = Q_1\overline{Q_0} + \overline{Q_1}\,\overline{x}\,Q_0$

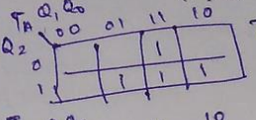
$Q_0^+ = \overline{Q_1}\,\overline{Q_0}\,\overline{x} + Q_1\overline{Q_0}\,x + Q_1Q_0\overline{x}$

Name: _____ Roll Number: _____

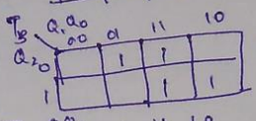
3. Design and draw the circuit of a synchronous mod-6 counter which counts from 000 in the binary sequence. Also, if any unused state occurs, the output goes to 000 at the next clock pulse. (4 pts)



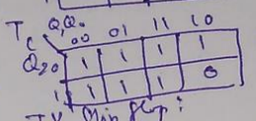
Present state			Next state			D-Flip-flop inputs			T-Flip flop inputs		
Q_2	Q_1	Q_0	Q_2	Q_1	Q_0	D_A	D_B	D_C	T_A	T_B	T_C
0	0	0	0	0	1	0	0	1	0	0	1
0	0	1	0	1	0	0	1	0	0	1	1
0	1	0	0	1	1	0	1	1	0	0	1
0	1	1	1	0	0	1	0	0	1	1	1
1	0	0	1	0	1	1	0	1	0	0	1
1	0	1	0	0	0	0	0	0	1	0	1
1	1	0	0	0	0	0	0	0	1	1	0
1	1	1	0	0	0	0	0	0	1	1	1



$T_A = A\bar{B}C + Q_2Q_1' + Q_1Q_0 + Q_2Q_0$



$T_B = Q_2'Q_0 + Q_2Q_1'$



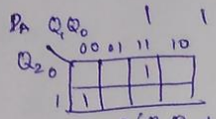
$T_C = Q_2' + Q_1' + Q_0$

JK flip flop:

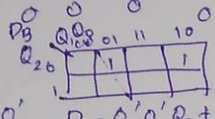
$J_A = Q_1'Q_0'$ $K_A = Q_1'Q_0'$

$J_B = Q_2'Q_0$ $K_B = Q_2'Q_0'$

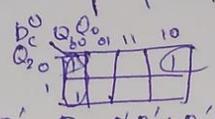
$J_C = Q_2Q_1'$ $K_C = 1$



$D_A = Q_2'Q_1Q_0 + Q_2Q_1'Q_0'$

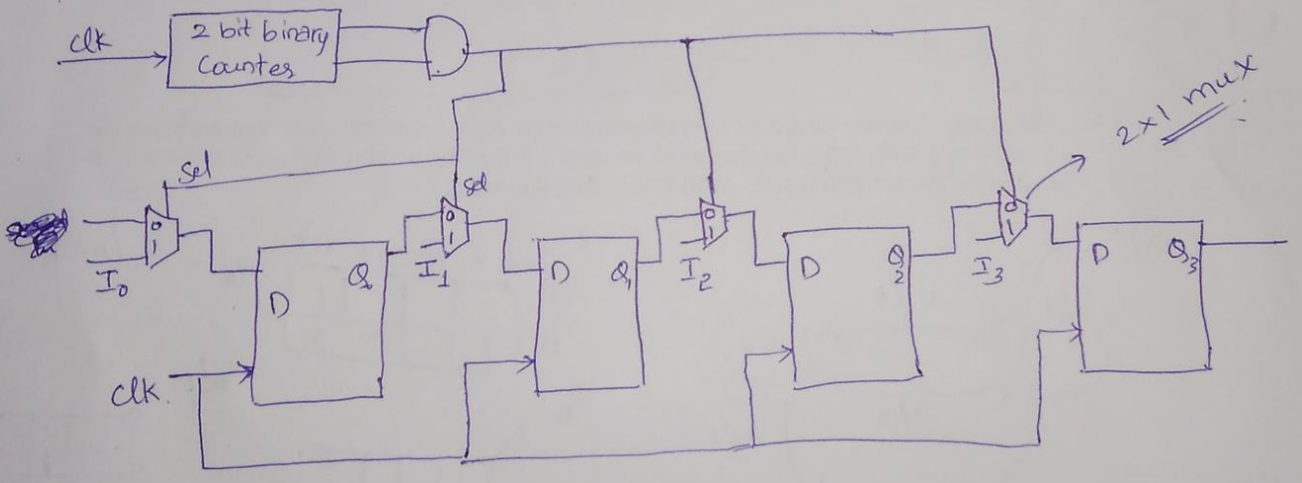


$D_B = Q_2'Q_1'Q_0 + Q_2'Q_1Q_0'$



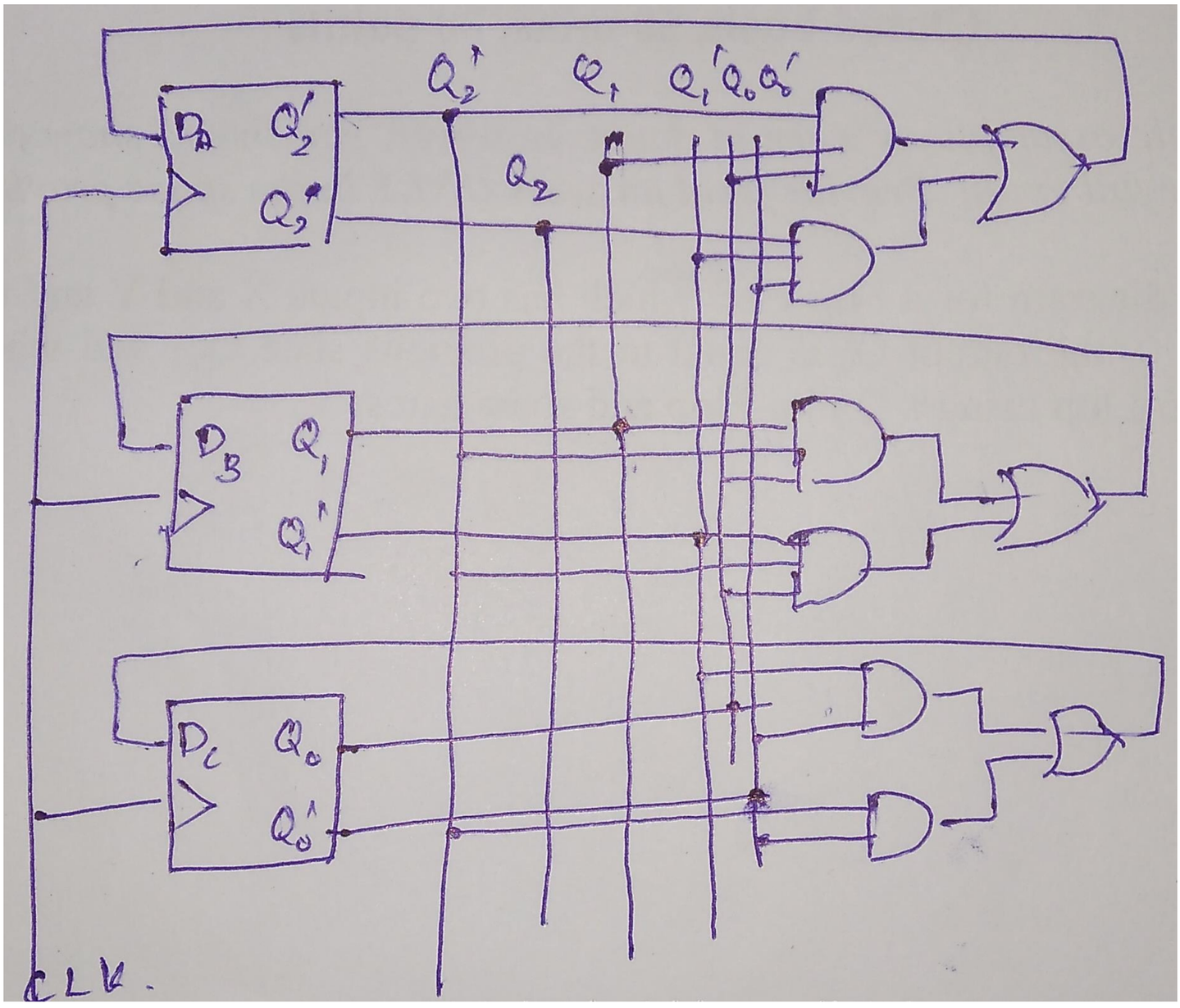
$D_C = Q_1'Q_0' + Q_2'Q_0'$

4. Draw the circuit of a system which loads 4-bit parallel data once in every 4 clock pulses and outputs them as 4-bit serial data. The design **must** include generation of appropriate control signals to load data at proper intervals. (4 pts)



5. (a) Consider a 16 M x 8 memory constructed using the above element. The number of address lines and the input/output data lines required are 24 and 8 respectively. (2pts)

(b) The number of AND gates required for the decoder in the above memory is 2^{24} for 1-D addressing and 2^{13} for 2-D addressing (2pts)


Q_2'
Q_1
$Q_1'Q_0Q_0'$
D_A
Q_2
Q_2'
Q_2
D_B
Q_1
Q_1'
D_C
Q_0
Q_0'
CLK.